

Materials Research Report; Kid's GoKart by Ryan Ferguson

Introduction:

This report is based on a kids pedal powered go-kart designed for mobility on smooth pavements. It consists of the body frame, four wheels, steering mechanism and a rear axle with a handbrake, connected from the pedals via a chain system. For this assignment, the focus will be on the material selection of the wheels and the rear axle. Currently, the material used for the wheels is PEEK (polyether ether ketone) and high alloy steel for the rear axle. Although these materials both provide high mechanical performance they both have significant material cost. When producing a batch order of 10,000 units, this has a severe impact on increasing production expense.

The purpose of this task is to identify more cost effective material alternatives while ensuring adequate mechanical performance and safety. Furthermore, this report will include a sustainability assessment on selected materials and processes. This is important because metrics such as energy consumption, CO₂ footprint and end of life considerations will be discussed and should positively benefit an appropriate selection. To address the cost reduction task while maintaining mechanical performance, I will be using Granta EduPack for a systematic materials and manufacturing selection approach. For the wheel, alternative polymer materials will be compared against PEEK to determine the limitations of minimum stiffness and yield strength requirements. For the rear axle, non-ferrous metals will most likely be the focus on a material alternative based on mechanical performance, cost and suitability for applications of loading. **Please note all table figures are maximum values.**

Materials Selection:

Wheel Material Selection:

The wheel with an integrated tire is a key component within the go-kart. It must vertically support the load of the rider, maintain stability and resist local stresses. The material must provide sufficient stiffness to prevent excessive deformation and appropriate yield strength to avoid failure under general surface loads. As stated in the design brief, the selected material must have a minimum Young's Modulus of 1 GPa and a minimum yield strength of 30 MPa. Additionally, the alternative material must offer a significant reduction in cost compared to PEEK.

Material research was undertaken in Granta Edupack using the Level 3 Materials Universe database.

Figure 1

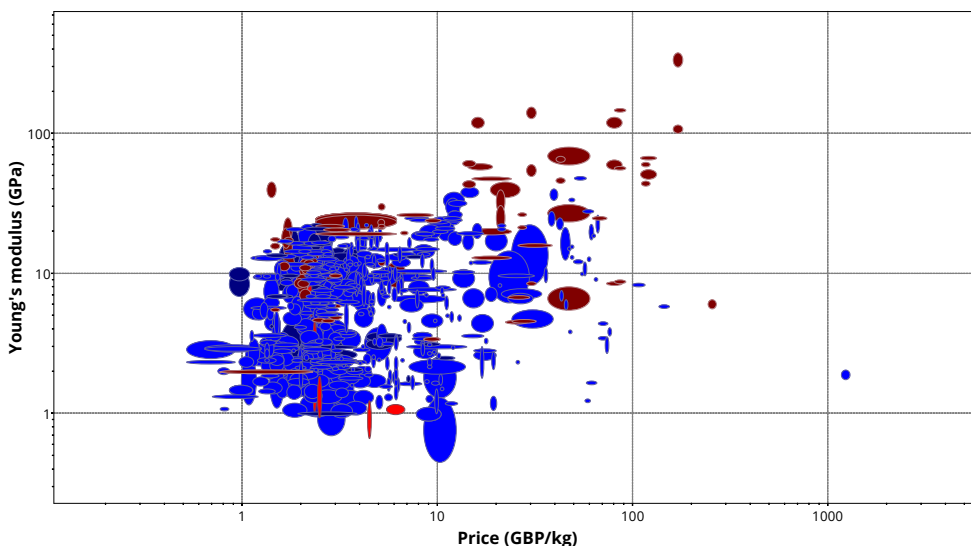


Figure 1 illustrates that by applying the limits of a minimum Young's Modulus of 1 GPa and a minimum yield strength of 30 MPa within the polymer plastics section, a total of 677 materials were included in the initial screening criteria. While this confirms that many polymers technically meet the minimum mechanical thresholds, the selection must be further refined based on cost and CO₂ footprint to identify the most suitable cost effective alternative to PEEK.

To further reduce the total of possible selected polymers, an additional price limit of £5/kg was applied. However, this still left just over 400 materials to choose from. In a new figure, I decided to refine details similar to PEEK but ensure there was a significant cost reduction and benefits within the material sustainability. The new limits on the chart included: A price limit of £5/kg, a maximum virgin grade production of 9kg/kg, a min Young's Modulus of 1 GPa and max of 2 GPa and a minimum yield strength of 30 MPa and max of 40MPa. A chart of Young's Modulus against price was then generated to visually identify materials offering the best combination of stiffness (very similar to PEEK) and affordability. There were three main material types repeated and identified that passed these limit stages: Polypropylene (PP), Acrylonitrile Butadiene Styrene (ABS), and Polyamide (Nylon). A second chart (Figure 2) comparing yield strength against price was produced to confirm all three choices satisfy the minimum yield strength requirement of 30 MPa. Table 1 summarises the key mechanical and cost properties of these candidates alongside PEEK for comparison.

Table 1

Material	Young's Modulus (GPa)*	Yield Strength (MPa)*	Price (£/kg)*	CO ₂ Footprint (kg/kg)* (Virgin grade)
PEEK (Current)	3.95	110	48.6	17.5
Polypropylene (PP) (30% talc, copolymer)	2.84	30	1.51	2.77
ABS (molding)	2.76	51	2.47	4.32
Nylon (PA6) (molding and extrusion)	1.18	48.2	2.14	8.48

ABS (molding) was selected as the preferred alternative material for the wheel component. With a Young's Modulus of 2.76 GPa and yield strength of 51 MPa, ABS satisfies both minimum mechanical requirements with considerable margin. Its price of £2.47/kg represents a cost reduction of approximately 94.9% to PEEK producing savings of approximately £461,300 across the target batch size of 10,000 units. Furthermore, its CO₂ footprint of 4.32 kg/kg virgin grade represents a 75.3% reduction compared to PEEK at 17.50 kg/kg, offering an improved environmental benefit. While PP (30% talc, copolymer) offered a marginally lower cost at £1.51/kg, its yield strength of exactly 30 MPa sits at the minimum permissible threshold, providing no safety margin. PA6 was rejected due to the elevated CO₂ footprint. ABS therefore represents a strong balance of mechanical performance, cost efficiency, and reduced environmental impact.

Figure 2

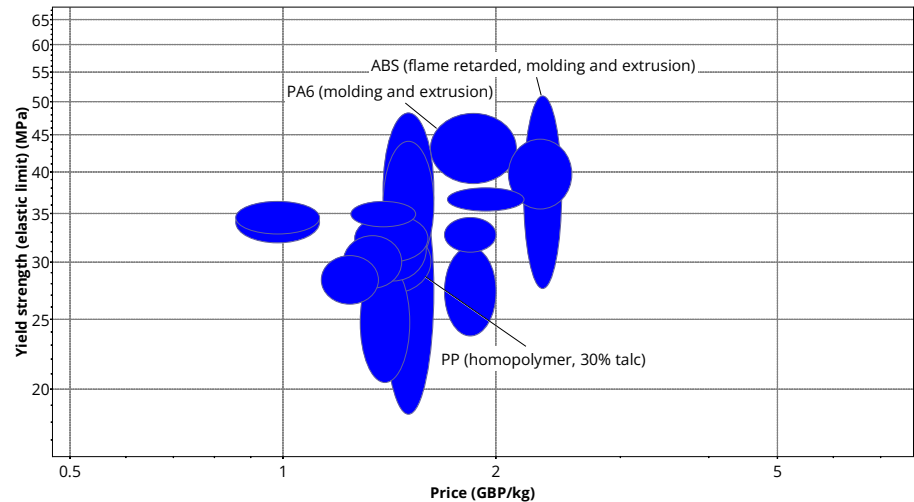


Figure 2 presents a chart of yield strength against price for the polymer selection. Three candidate materials are identifiable within the selection region: ABS (flame retarded, molding and extrusion), PA6 (molding and extrusion), and PP (homopolymer, 30% talc). All three candidates satisfy the minimum yield strength threshold. However, ABS and PA6 sit comfortably above the 30 MPa minimum, while PP (30% talc) sits at approximately the threshold boundary, offering minimal safety margin. Both ABS and PA6 remain within the low-price region of under £5/kg, confirming their cost advantage over PEEK at £48.60/kg

Rear Axle Material Selection:

The rear axle must transmit torque from the pedal-chain system to the wheels whilst resisting bending loads. As specified in the design brief, the alternative material must be selected from the **non-ferrous** metals category with a minimum Young's Modulus of 50 GPa. Material research was undertaken in Granta EduPack using the Level 3 Materials Universe database, filtering by the non-ferrous metals family with the Young's Modulus limit applied.

Figure 3

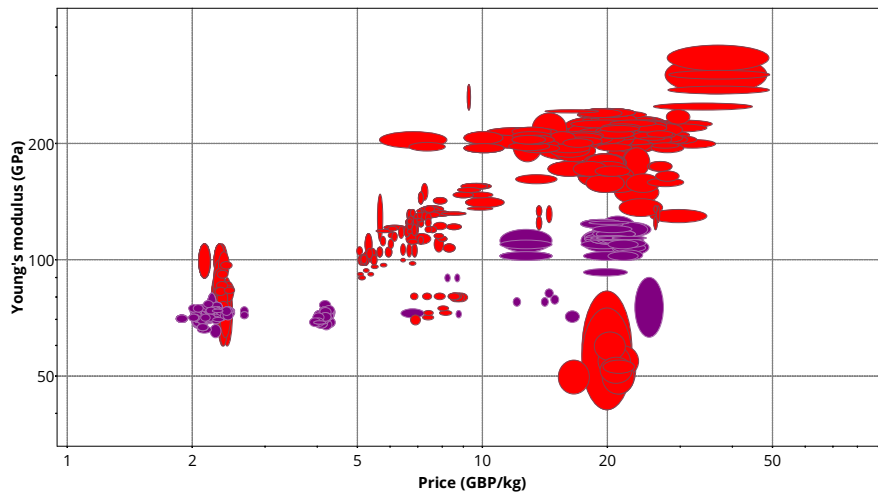
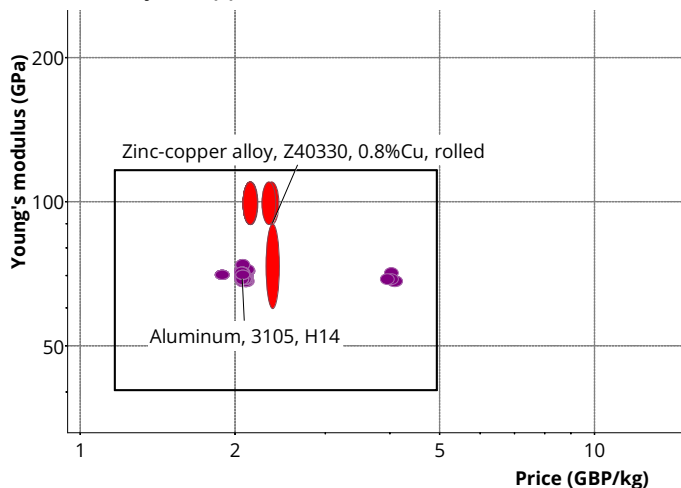


Figure 3 presents a bubble chart of Young's Modulus against price for non-ferrous metal selection, following the application of limits previously mentioned. The constraints applied include a minimum Young's Modulus of 50 GPa, a non-ferrous metals filter, and I thought it would be a good idea to include a maximum price limit of £28.20/kg equivalent to the current cost of high alloy steel ensuring any selected alternative offers the additional cost benefit.

To further refine my selection for the rear axle, I have decided to focus on sustainability factors. This means setting a limit on materials that can be recycled both naturally and functionally. This ensures that the selected material not only meets mechanical and cost requirements but also supports responsible end-of-life management. Following the application of this filter alongside the existing constraints of minimum Young's Modulus of 50 GPa and price limit of £28.20/kg, the selection pool narrowed considerably. The remaining materials were predominantly aluminium alloy and zinc alloy materials, both of which are commonly used in lightweight components such as axles, hubs and bicycle applications.



Selected candidates and why:

- Aluminium 3105, H14 - Good general purpose alloy
- Aluminium 5052, H32 - Stronger than 3105, excellent corrosion resistance, very common
- Aluminium 5083, H321 - Higher strength structural grade, good for higher loads
- Zinc-copper alloy Z40330 - Represents the zinc material for comparison

Furthermore, all of these materials are all recyclable making them an appropriate selection.

Table 2

Material	Young's Modulus (GPa)*	Yield Strength (MPa)*	Price (£/kg)*	CO ₂ Footprint (kg/kg)* (Virgin grade)
Aluminium 3105	72	158	2.13	13.03
Aluminium 5052	73.6	172	2.17	13.6
Aluminium 5083	73.6	221	2.17	13.7
Z40330	90	185	2.43	3.86

Four candidates were selected for detailed comparison. Aluminium 3105 H14 was included as a representative general purpose aluminium alloy with moderate strength and low cost. Aluminium 5052 H32 was selected due to its improved yield strength relative to 3105 while remaining within a similar price range, making it a stronger mechanical candidate. Aluminium 5083 H321 was included as a higher strength structural grade, commonly used in load applications, providing an improved performance advantage within the aluminium family. Finally, Zinc-copper alloy Z40330 was included to represent the zinc alloy family identified in the earlier screening stage, allowing a fair material on mechanical performance.

Following comparison of the four shortlisted candidates in Table 2, Aluminium 5052 was selected as the preferred alternative material for the rear axle. All three aluminium alloys satisfy the minimum Young's Modulus requirement of 50 GPa with considerable margin, ranging from 72 to 73.6 GPa. Aluminium 5052 was preferred over 3105 due to its higher yield strength of 172 MPa compared to 158 MPa, providing greater resistance to the bending and torsional loads experienced during pedal operation. While Aluminium 5083 offered the highest yield strength at 221 MPa, the slight improvement over 5052 did not justify its higher CO₂ footprint of 13.7 kg/kg. Z40330 zinc-copper alloy was rejected despite meeting the stiffness requirement, as its significantly higher density of 7.18 g/cm³ compared to aluminium's 2.7 g/cm³ results in a considerably heavier axle, increasing the pedalling effort required from the child rider. At £2.17/kg, Aluminium 5052 represents a cost reduction of approximately 92.3% compared to high alloy steel at £28.20/kg, producing excellent savings across the 10,000 unit batch while maintaining sufficient mechanical performance and improved sustainability factors.

Manufacturing Process Selection

Wheel Manufacturing:

For the wheel component, manufacturing process selection was carried out using the Process Universe database in Granta EduPack. The key requirements applied reflect the design brief guide: a primary discrete process class, a 3D solid shape class, a component mass of 1 kg, a minimum section thickness of 4 mm, and a batch size of 10,000 units. Following the application of these filters, the results were narrowed to a shortlist of 27 candidate processes for further evaluation.

Name	
BMC molding	Hot closed die forging
Centrifugal casting	Hot isostatic pressing (HIP), stan...
Centrifugally-aided casting	Hot open die forging
Cold closed die forging	Injection molding (thermoplastics)
Cold isostatic pressing (CIP)	Injection molding (thermosets)
Compression molding	Reaction injection molding
Cosworth casting	Replicast casting
Die pressing and sintering	Rheocasting
Evaporative pattern casting, aut...	Shell casting
Expanded foam molding	SMC molding
Ferro die casting	Squeeze casting
Gravity die casting	Thixocasting
Green sand casting, automated	Transfer molding
High pressure die casting	

By then comparing this list to processes that are compatible with ABS links for processes and aiming to use molding as my manufacturing technique, two processes were compared.

Table 3

Process	Tooling Cost	Capital Cost	Labour Intensity	Relative Cost Index
Injection Molding	7.21e4	6.49e5	Low	85.4
Compression Molding	1.44e4	8.66e4	Low	32.9

Table 3 presents a comparison of the relative costs for the two primary processes obtained from Granta EduPack process datasheets. Injection moulding, despite having the higher maximum costs, distributes this expense across 10,000 units resulting in the lowest cost per unit at the target batch size, making it the most economically viable process for this production volume. Compression molding, whilst cheaper in tooling, is better suited to lower batch sizes and thermoset materials, limiting its suitability for ABS at this production volume. **Therefore I have chosen injection molding for my manufacturing process mainly due to the following:**

- **Material compatibility:** EduPack specifically categorises injection moulding for thermoplastics as a distinct process, directly confirming compatibility with ABS which is a thermoplastic.
- **Geometry suitability** - injection moulding is capable of producing complex 3D solid geometries with features such wheel face and tyre geometry in a single operation, ensuring the wheel's shape class requirement.
- **Batch size efficiency** - At a production volume of 10,000 units, the high tooling cost of £72,100 is effectively repaid in the batch, yielding a low per-unit cost that alternative processes cannot match at this scale.
- **Labour intensity** - As confirmed by the EduPack datasheet, labour intensity is rated as low, reducing overall production costs at scale compared to more manually intensive processes.
- **Process of elimination** - Whilst other processes such as gravity die casting, high pressure die casting and green sand casting appeared in the refined list, these are metal forming processes incompatible with ABS.

Rear Axle Manufacturing:

For the rear axle, manufacturing process selection was carried out with similar limits. A circular prismatic shape class was selected to represent the cylindrical geometry of the rear axle. A mass range of 1 kg was set for both the minimum and maximum values, and a minimum section thickness of 4 mm was applied. The process characteristic was set to discrete, reflecting the requirement for a primary discrete manufacturing process. Finally, an economic batch size of 10,000 units was applied to both the minimum and maximum ensuring only processes economically viable at the target production volume were considered. These produced 37 process stages resulting in a pass.

Name	
Abrasive jet machining (AJM)	Hot closed die forging
Broaching	Hot open die forging
Centerless grinding	Hot pressing
Centrifugal casting	Impact extrusion
Centrifugal molding	Injection molding (thermoplastics)
Centrifugally-aided casting	Injection molding (thermosets)
Cold closed die forging	Milling
Cold isostatic pressing (CIP)	Reaction injection molding
Cosworth casting	Replicast casting
Die pressing and sintering	Repressing
Drilling	Rheocasting
Evaporative pattern casting, aut...	Shell casting
Expanded foam molding	Spark sintering
Ferro die casting	Squeeze casting
Filament winding	Swaging
Gravity die casting	Thixocasting
Green sand casting, automated	Transfer molding
Grinding	Turning, boring and parting
High pressure die casting	

The process list was refined to identify suitable candidates for producing the rear axle. Comparing the following results compatible with Aluminium 5052, three processes were shortlisted for detailed comparison: impact extrusion, cold closed die forging, and drawing. Impact extrusion was identified as the primary candidate due to its suitability for producing cylindrical aluminium components at high volume. Hot closed die forging was included as a high-performance alternative given its ability to produce better mechanical properties through grain refinement. Cold closed die forging was considered as a similar but lower-cost process.

Within the datasheets of these three methods, **impact extrusion seems to be the most suitable and my chosen process due to the following reasons:**

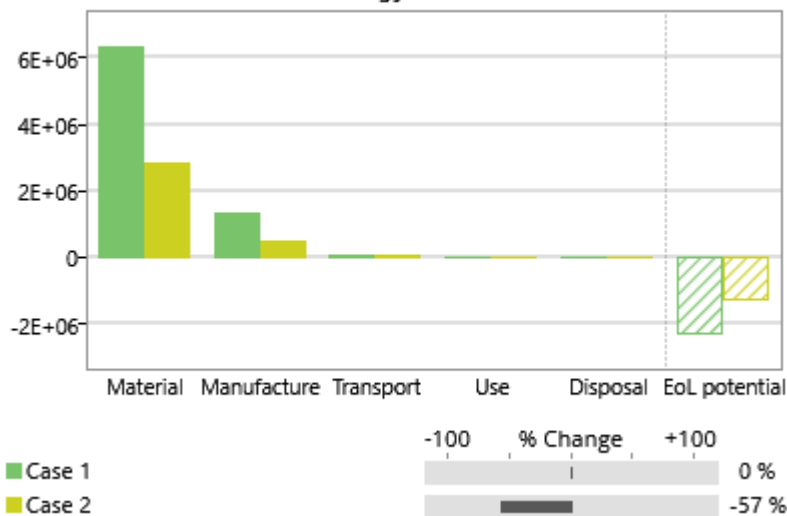
- **Cost effectiveness** - Only £15 per unit compared to a high £27 and £45 for cold and hot die forging.
- **Material compatibility** - Aluminium alloys are exceptionally well suited to impact extrusion due to their high ductility.
- **Mass production suitability** - impact extrusion is specifically designed for mass production suitable for 10,000 unit requirement.

Sustainability Assessment:

A sustainability assessment was carried out using the Eco Audit tool in Granta EduPack, comparing the environmental impact of the original materials (PEEK for the wheel and high alloy steel for the rear axle against the selected alternatives of ABS and Aluminium 5052 respectively). Both audits were configured with identical component masses of 1 kg, manufacturing processes, transport assumptions, and end-of-life scenarios to ensure a controlled comparison. The assessment focused on three key metrics: energy consumption, CO₂ emissions, and cost, the results of which are presented in Figures 5, 6, and 7 respectively.

Figure 5

Energy (MJ)



Overall, the Eco Audit results indicate a **total energy reduction of 57%** when switching from the original to the alternative materials, demonstrating a substantial improvement in energy efficiency across the product lifecycle.

Figure 6

Climate change (CO₂-eq) (kg)

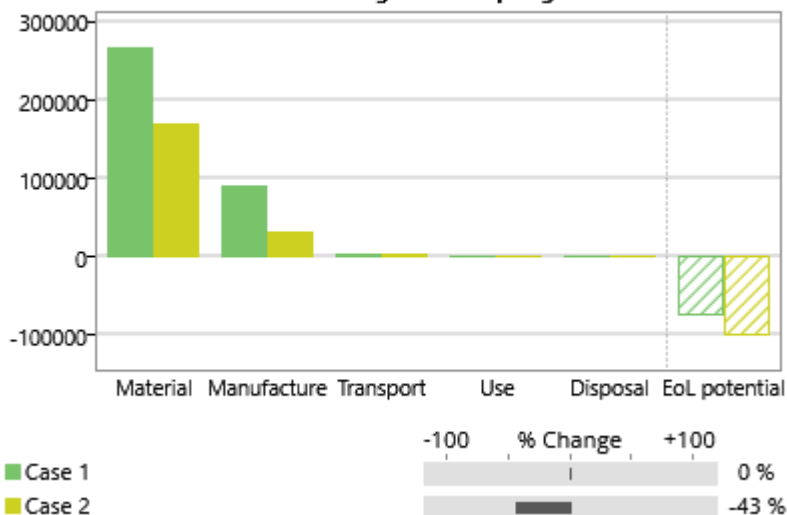
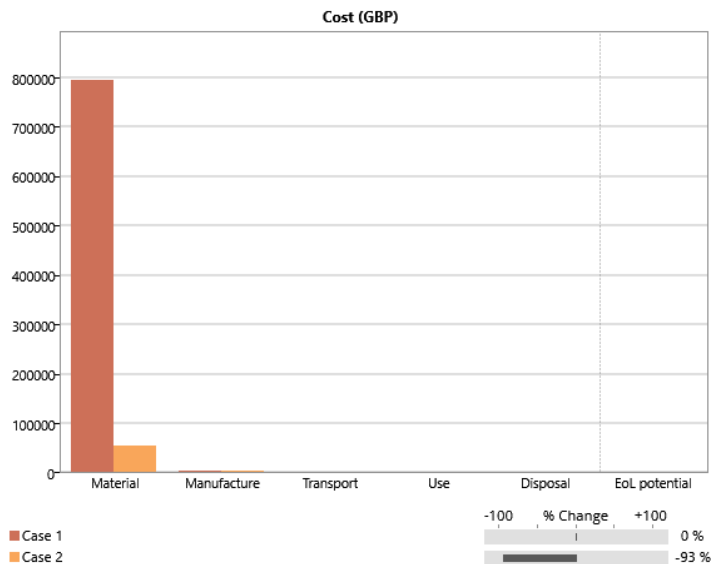


Figure 5 presents the energy consumption comparison between Case 1 (original materials) and Case 2 (alternative materials) across all lifecycle phases for a batch of 10,000 units. The material production phase is the clear contributor to energy consumption in both cases, significantly outweighing the manufacture, transport, use and disposal phases. This is consistent with expectations for high energy virgin grade material production. The material phase energy reduction is approximately 50%. The manufacturing phase also shows a modest reduction in Case 2, reflecting the lower processing temperatures required for ABS compared to PEEK, which requires elevated temperatures due to its better structure. The use phase contributes negligible energy in both cases, which is expected given the human powered function of the go-kart. The end-of-life potential, shown as negative hatched bars, indicates energy recovery through recycling for both cases, with Case 2 demonstrating marginally greater recovery potential attributable to the high recyclability of Aluminium 5052.

Figure 6 presents the CO₂ emissions comparison between Case 1 and Case 2. As with the energy assessment, the material production phase represents the main source of CO₂ emissions in both cases. Case 1 records a material phase CO₂ contribution of approximately 270,000 kg compared to approximately 170,000 kg for Case 2, reflecting the significantly higher CO₂ footprint of PEEK at 17.5 kg/kg virgin grade relative to ABS at 4.32 kg/kg.

The manufacturing phase also shows a considerable reduction in Case 2, from approximately 90,000 kg to 30,000 kg. The end-of-life potential bars indicate a greater CO₂ recycling benefit for Case 2, expected with the high recyclability of Aluminium 5052. **The total CO₂ reduction achieved by switching to the alternative materials is 43%**, representing a substantial improvement in the carbon footprint of the go-kart across its production lifecycle.

Figure 7



The cost graph provides clear evidence of the benefit of the alternative material selection. Case 1 records a total material cost of approximately £800,000 across the batch, driven predominantly by the high unit price of PEEK at £48.60/kg for the wheel component. In contrast, Case 2 records a material cost of approximately £55,000, representing an overall cost reduction of 93% as indicated by the Eco Audit percentage change. This finding directly justifies the aim of this assignment, which was to identify cost-effective material alternatives without compromising mechanical performance. The major cost reduction demonstrated in Figure 7, combined with the 57% energy reduction and 43% CO₂ reduction shown in Figures 5 and 6 illustrate that the alternative material selections of ABS and Aluminium 5052 deliver great improvements across all three sustainability and economic metrics evaluated using the Eco Audit tool.

Conclusion:

Chosen Materials and Why?

Wheel – ABS replacing PEEK

- Young's Modulus of 2.76 GPa - exceeds 1 GPa minimum requirement
- Yield strength of 51 MPa - exceeds 30 MPa minimum with safe margin
- 94.9% cost reduction - from £48.60/kg down to £2.47/kg
- 75.3% reduction in CO₂ footprint - from 17.5 to 4.32 kg/kg virgin grade
- Mechanically recyclable - improved end of life sustainability over PEEK
- Fully compatible with injection molding at mass production volumes

Rear Axle – Aluminium 5052 H32 replacing High Alloy Steel

- Young's Modulus of 73.6 GPa - comfortably exceeds 50 GPa minimum requirement
- Yield strength of 172 MPa - strong resistance to axle bending and loads
- 92.3% cost reduction - from £28.20/kg down to £2.17/kg
- Non-ferrous - satisfies requirement
- Density of ~2.7 g/cm³ - significantly lighter than steel
- 90% recyclability rate - one of the most recycled metals
- Excellent compatibility with impact extrusion for mass production of cylindrical components

This report has identified cost-effective material and manufacturing process alternatives for the wheel and rear axle of a child's pedal go-kart using Granta EduPack. ABS (molding) was selected to replace PEEK for the wheel, satisfying all minimum mechanical requirements whilst achieving a 94.9% cost reduction. Aluminium 5052 H32 was selected to replace high alloy steel for the rear axle, exceeding the minimum stiffness requirement whilst delivering a 92.3% cost reduction. Injection molding and impact extrusion were identified as the most suitable manufacturing processes for the wheel and axle respectively, based on material compatibility, geometry, and cost effectiveness at the target batch size of 10,000 units.

The Eco Audit confirmed that the alternative selections deliver a 57% reduction in energy consumption, 43% reduction in CO₂ emissions, and 93% reduction in material cost across the production batch, with material production identified as the main lifecycle phase in all three metrics.